Physics 438B – Chapter #12
Charged Particle in a Magnetic Field

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1 Classical Electromagnetism

We begin with a brief review of the classical description of a charged particle in an electromagnetic field. While these ideas are developed more fully in courses such as Physics 408, only a limited subset will be needed here. In particular, we introduce the scalar and vector potentials, which provide the most general description of particle-field interactions consistent with Galilean relativity (we encountered this idea in Chapter #11).

Some of the formalism—especially in its Lagrangian and Hamiltonian form—may be unfamiliar. However, the essential results can be followed without a comprehensive background in electromagnetism. The goal of this review is to highlight that several features which may appear distinctly quantum, such as the significance of potentials, already arise at the classical level. A working knowledge of classical mechanics, including Lagrangian and Hamiltonian formulations, will be assumed.

The electric and magnetic fields, $\mathbf{E}$ and $\mathbf{B}$, enter the Lagrangian formulation of classical mechanics through the scalar and vector potentials, $\mathbf{A}$ and ϕ, defined such that

$$\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t} \tag{1}$$

$$\mathbf{B} = \nabla \times \mathbf{A} \tag{2}$$

In classical E&M, the fields are considered real physical fields defined at every point in space and time. Operationally, they are defined by the forces they produce on charges, and dynamically, they are independent degrees of freedom governed by Maxwell's equations. They can exist and propagate even in the absence of sources, carry energy and momentum, and interact with charges and currents. The potentials that are used to describe the fields are not considered physical; they are viewed as auxiliary quantities that are useful within a mathematical framework, but do not themselves represent any observable features of nature.

Within this classical framework, the potentials are not unique. The fields $\mathbf{E}$ and $\mathbf{B}$ are unaffected by the following transformation:

$$\mathbf{A} \mapsto \mathbf{A}' = \mathbf{A} + \nabla\Lambda, \quad \phi \mapsto \phi' = \phi - \frac{\partial \Lambda}{\partial t} \tag{3}$$

where Λ is an arbitrary real scalar function of position and time. Eq. (3) is called a **gauge transformation** and the theory is called **gauge invariant**. In particular, the fields are called gauge invariant quantities under this transformation. Informally, gauge invariance reflects a redundancy in how we describe the system, not a symmetry that changes its physical state. The "extra freedom" only concerns the mathematical representation and indicates that our variables contain more information than the physics actually requires.

The Lagrangian[1] for a charged particle of mass m and charge q in an arbitrary electromagnetic field is given by

$$L(\mathbf{r}, \mathbf{v}, t) = \tfrac{1}{2} m \mathbf{v}^2 + q \mathbf{v} \cdot \mathbf{A}(\mathbf{r}, t) - q\phi(\mathbf{r}, t) \tag{4}$$

where $\mathbf{r}$ and $\mathbf{v}$ are the position and velocity of the particle. The Euler-Lagrange equations,

$$\frac{d}{dt}\left[\frac{\partial L}{\partial v_\alpha}\right] - \frac{\partial L}{\partial r_\alpha} = 0, \quad (\alpha = 1, 2, 3), \tag{5}$$

lead to the correct Newtonian equation of motion in terms of the Lorentz force law:

$$m\frac{d\mathbf{v}}{dt} = q\left(\mathbf{E} + \mathbf{v} \times \mathbf{B}\right) \tag{6}$$

From the Lagrangian, we can calculate the *canonical momentum* for a particle in an electromagnetic field:

$$p_\alpha = \frac{\partial L}{\partial v_\alpha} \;\Rightarrow\; \mathbf{p} = m\mathbf{v} + q\mathbf{A} \tag{7}$$

Notice that $\mathbf{p}$ changes under the gauge transformation given by Eq. (3), and hence lacks direct physical significance. Its importance is primarily mathematical, arising from its conservation when the system is invariant under spatial translations. More precisely, when the Lagrangian is invariant under spatial translation, the canonical momentum is conserved, and not the more intuitive quantity $m\mathbf{v}$.

The classical Hamiltonian for a particle in an electromagnetic field is given by

$$H = \mathbf{v} \cdot \mathbf{p} - L = \frac{1}{2m}\left(\mathbf{p} - q\mathbf{A}\right)^2 + q\phi \tag{8}$$

Since the first term is really just $\frac{1}{2}mv^2$, the Hamiltonian corresponds to the total energy: the sum of the kinetic energy and the electric potential energy. Hamilton's equations yield the familiar Newtonian equation of motion.

[1] See Taylor, *Classical Mechanics*, 2005.

2 Quantum Theory

The most general Hamiltonian for a spinless particle of mass m that respects the Galilean symmetries of spacetime is given by

$$H = \frac{1}{2m}\left(\mathbf{p} - \alpha\mathbf{A}\right)^2 + \beta\phi$$

where $\mathbf{A}$ is a vector potential, ϕ is a scalar potential, and the scalar constants α and β are included to account for the observation that particles with the same mass may move differently in a region of space containing the same $\mathbf{A}$ and ϕ, possibly as a result of some other physical characteristic (e.g., the charge). In a nonrelativistic theory, there doesn't seem to be any clear arguments that $\alpha = \beta = q$, the charge of the particle. For simplicity, we'll take for granted that the coupling of the particle to the electromagnetic field is proportional to the particle's charge q and rewrite the Hamiltonian in the more familiar form

$$H = \frac{1}{2m}\left(\mathbf{p} - q\mathbf{A}\right)^2 + q\phi \tag{9}$$

Heisenberg Picture & Gauge Transformation

For simplicity of notation we shall not distinguish between the Schrödinger and Heisenberg operators (see Chapter #3 for a discussion of the Heisenberg picture). In the Heisenberg picture, the rate of change of the position operator is

$$\frac{d\mathbf{r}}{dt} = \frac{i}{\hbar}[H, \mathbf{r}] = \frac{\mathbf{p} - q\mathbf{A}}{m} \tag{10}$$

which is just the velocity operator $\mathbf{v}$. Since expectation values of $\mathbf{v}$ are tied to measurable changes in the expectation values of $\mathbf{r}$, we regard the velocity operator as more *physically relevant* than either $\mathbf{p}$ or $\mathbf{A}$. The Heisenberg equation of motion for $m\mathbf{v}$ is

$$m\frac{d\mathbf{v}}{dt} = \frac{im}{\hbar}[H, \mathbf{v}] + m\frac{\partial \mathbf{v}}{\partial t} \tag{11}$$

and after some tedious calculations, we ultimately find

$$m\frac{d\mathbf{v}}{dt} = -q\nabla\phi - q\,\frac{\partial \mathbf{A}}{\partial t} + \frac{q}{2}\Big[\mathbf{v} \times (\nabla \times \mathbf{A}) - (\nabla \times \mathbf{A}) \times \mathbf{v}\Big] \tag{12}$$

The operators $\mathbf{E} = -\nabla\phi - \partial_t \mathbf{A}$ and $\mathbf{B} = \nabla \times \mathbf{A}$ are tied to measurable changes in $\mathbf{v}$, so we regard them as physically relevant. The field operators are invariant under

$$\mathbf{A} \mapsto \mathbf{A}' = \mathbf{A} + \nabla\Lambda, \quad \phi \mapsto \phi' = \phi - \frac{\partial \Lambda}{\partial t} \tag{13}$$

and furthermore, the Schrödinger equation in the coordinate representation is invariant under the transformation above as long as we include

$$\psi \mapsto \psi' = e^{i(q/\hbar)\Lambda}\psi \tag{14}$$

which, as we know, leaves the state of the system unchanged, e.g., $|\psi|^2$ is invariant.

Although it appears the velocity operator is gauge dependent, *its expectation value remains unchanged.* The expectation value of momentum, however, is not gauge invariant. For this reason, the physical significance of a result will usually be more apparent if it is expressed in terms of the velocity, rather than in terms of the momentum. It is therefore worth examining the mathematical properties of the velocity operator. For instance, consider the commutator of the velocity components with each other:

$$[v_x, v_y] = -\frac{q}{m^2}\Big([A_x, p_y] + [p_x, A_y]\Big)$$

Consider an arbitrary position-space wave function $\psi(\mathbf{r})$:

$$\begin{aligned}
[v_x, v_y]\psi &= \frac{i\hbar q}{m^2}\left(A_x \frac{\partial}{\partial y} - \frac{\partial}{\partial y} A_x + \frac{\partial}{\partial x} A_y - A_y \frac{\partial}{\partial x}\right)\psi \\
&= \frac{i\hbar q}{m^2}\left(-\frac{\partial A_x}{\partial y} + \frac{\partial A_y}{\partial x}\right)\psi \\
&= \frac{i\hbar q}{m^2} B_z \psi
\end{aligned}$$

Since this result holds for an arbitrary wave function, we can write $[v_x, v_y] = (i\hbar q/m^2)B_z$, which may be generalized to

$$[v_i, v_j] = \frac{i\hbar q}{m^2}\epsilon_{ijk} B_k \tag{15}$$

In words, the commutator of two components of velocity is proportional to the magnetic field in the remaining direction. If the magnetic field is zero in any particular direction, the velocity components perpendicular to that direction will commute with one another, and thus will share a common set of eigenstates.

3 Motion in a Uniform Magnetic Field

Consider a charged particle moving in a region of space where all field components are zero except B_z, which is a multiple of the identity. Although many choices for the potentials are consistent with this arrangement, we will assume the vector potential is static and the scalar potential vanishes. Since B_x and B_y are both zero, it follows that v_z commutes with v_x and v_y. The Hamiltonian is

$$H = \frac{1}{2}m(v_x^2 + v_y^2) + \frac{1}{2}mv_z^2 = H_{xy} + H_z \tag{16}$$

where the operators $H_{xy} = \frac{1}{2}m(v_x^2 + v_y^2)$ and $H_z = \frac{1}{2}mv_z^2$ commute with each other, and thus have common eigenstates. Every eigenvalue of H is just the sum of an eigenvalue of H_{xy} and an eigenvalue of H_z.

We can derive the energy levels by first making the follow substitutions:

$$v_x = \frac{\sqrt{\hbar |q| B_z}}{m}\, Q, \quad \text{and} \quad v_y = \frac{\sqrt{\hbar |q| B_z}}{m} \frac{q}{|q|}\, P \tag{17}$$

From the commutation relation for the velocity components, it follows that $[Q, P] = i$ and the transverse Hamiltonian becomes

$$H_{xy} = \frac{\hbar |q| B_z}{m} \frac{1}{2}\Big(Q^2 + P^2\Big) \tag{18}$$

which is structurally identical to the Hamiltonian for a harmonic oscillator! The eigenvalues of the transverse Hamiltonian are therefore given by

$$E_{xy} = \Big(n + \tfrac{1}{2}\Big)\hbar\omega_c \tag{19}$$

where n is a nonnegative integer and $\omega_c = |q|B_z/m$ is the **cyclotron frequency**. The eigenvalue spectrum of H_z is continuous from $-\infty$ to ∞, so the energy eigenvalues for a charged particle in a uniform static magnetic field are

$$E_n(v_z) = \Big(n + \tfrac{1}{2}\Big)\hbar\omega_c + \tfrac{1}{2}mv_z^2 \tag{20}$$

The motion parallel to the magnetic field is not coupled to the transverse motion, and is unaffected by the field. The classical motion in the plane perpendicular to the field is in a circular orbit with angular frequency ω_c. Periodic motions correspond to discrete energy levels whose separation is $\hbar\omega_c$.

Energy Eigenfunctions

Although we were primarily interested in the energy eigenvalues, we can also determine the energy eigenfunctions. Let's ignore the motion in the z direction and confine our attention to the transverse motion. There are a few choices for the vector potential; let's go with $A_y = xB_z$. This gives the correct magnetic field with only a z component B_z. The Hamiltonian,

$$H = \frac{p_x^2}{2m} + \frac{1}{2m}\Big(p_y - xqB_z\Big)^2 \tag{21}$$

clearly commutes with p_y, but not with p_x due to the presence of x. We can therefore search for common eigenstates of H and p_y, with the latter being plane waves proportional to $e^{ik_y y}$ where $\hbar k_y$ is the momentum eigenvalue. In the coordinate representation, the energy eigenvalue equation $H\psi = E\psi$ is

$$\left[-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) + \frac{q^2B_z^2}{2m}\,x^2 + i\hbar\frac{qB_z}{m}\,x\,\frac{\partial}{\partial y}\right]\psi = E\psi \tag{22}$$

Since ψ is a common eigenstate of H and p_y, we will substitute

$$\psi(x,y) = e^{ik_y y}\phi(x) \tag{23}$$

and reduce the partial differential equation to an ordinary differential equation

$$\left[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{\hbar^2k_y^2}{2m} + \frac{q^2B_z^2}{2m}x^2 - \hbar k_y\frac{qB_z}{m}\,x\right]\phi = E\phi \tag{24}$$

The second term in the square brackets above can be written as a complete square by defining $x_0 = \hbar k_y/qB_z$. The differential equation then takes the form

$$\left[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2}m\omega_c^2(x-x_0)^2\right]\phi = E\phi \tag{25}$$

which is, unsurprisingly, exactly the differential equation for a simple harmonic oscillator. The function $\phi(x)$ is a harmonic oscillator eigenfunction, and the complete energy eigenfunction for the transverse Hamiltonian is

$$\psi(x,y) = H_n\left(\sqrt{\frac{m\omega_c}{\hbar}}\,(x-x_0)\right)\exp\left(-\frac{m\omega_c}{2\hbar}(x-x_0)^2 + ik_y y\right) \tag{26}$$

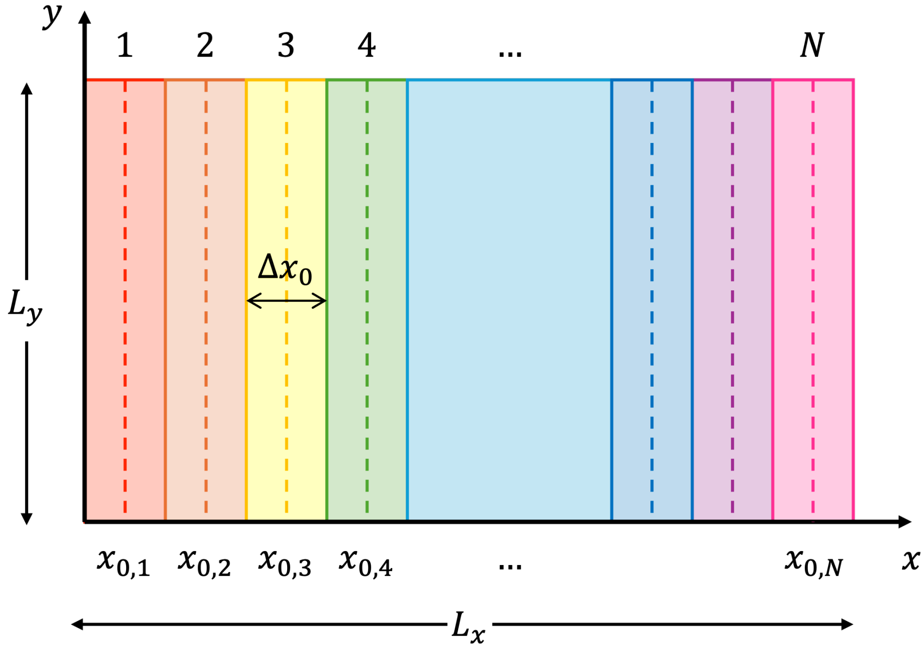

Figure 3.1: Representation of *Landau levels* in the x–y plane for a region of size $L_x \times L_y$. Each vertical band (parallel to the y-axis) corresponds to a quantum state labeled by its guiding center x_0, with states extended along y and localized in x about x_0, and adjacent centers separated by Δx_0. The states are indexed by $k_y = 2\pi j/L_y$ with $j \in \mathbb{Z}$, where $\pm j$ correspond to plane waves propagating in the $\pm y$ directions and $j = 0$ to a non-propagating state, so the total number of states is approximately twice the number of positive j values (plus one for $j = 0$). Each band occupies an effective area $\Delta A = L_y \Delta x_0 = \Phi_0/B_z$, corresponding to one flux quantum, and the total number of such bands within $0 \le x \le L_x$ is proportional to the Landau level degeneracy $N \propto \Phi/\Phi_0$.

Degeneracy

Since the energy depends only on n, there is degeneracy in the energy eigenstates corresponding to possible values of k_y. For each fixed n, there is a family of states labeled by k_y with the same energy. Different values of k_y correspond to different values of x_0, the center of the effective harmonic oscillator potential. Therefore, each eigenstate corresponds to the same cyclotron motion, but with a different guiding center position. The magnetic field fixes the energy of the orbit, but not where the orbit sits in space.

If the space available to the particle is infinite, then k_y is continuous and there are infinitely many possible values of x_0. Let's put the particle in a box of size $L_x \times L_y$. With the gauge potential $A_y = xB_z$, the Hamiltonian is translationally invariant in the y direction, so it is natural to impose periodic boundary conditions there, i.e., $\psi(x, y + L_y) = \psi(x, y)$.

This condition leads to

$$k_y = \frac{2\pi j}{L_y}, \quad j \in \mathbb{Z}$$

Since the wave function is localized around the point x_0, it is natural to confine the center to the region $0 \leq x_0 \leq L_x$. We'll assume the orbits are small compared to the overall size of the box so we can ignore the small fraction of those states lying close to the boundaries. In this limit, the number of degenerate states is approximately

$$2j_{\max} = 2\ \frac{|q|B_z}{2\pi\hbar} L_x L_y = 2\ \frac{\Phi}{\Phi_0} \tag{27}$$

where $\Phi = B_z L_x L_y$ is the *total magnetic flux* passing through the region $L_x \times L_y$ and $\Phi_0 = 2\pi\hbar/|q|$ is the *magnetic flux quantum.* There is a wonderful geometrical interpretation: each state is associated with an area of magnitude Φ_0/B_z in the plane perpendicular to the magnetic field. The quantity Φ_0 is a natural unit of magnetic flux. Thus the degeneracy factor of an energy level, called a Landau level in this context, is simply equal to the number of units of magnetic flux passing through the system.

4 Aharonov–Bohm Effect

Suppose a particle is moving in a region of space with a time-independent vector potential $\mathbf{A}$ *and* $\mathbf{B} = 0$. There are situations where $\mathbf{A}$ may be nonzero but we can effectively remove it from the quantum description with a gauge transformation. In the coordinate representation, the Schrödinger equation is

$$i\hbar\frac{\partial\psi}{\partial t} = \left[\frac{1}{2m}\left(-i\hbar\nabla - q\mathbf{A}\right)^2\right]\psi$$

We can simplify the Schrödinger equation with the following gauge transformation:

$$\mathbf{A} \mapsto \mathbf{A}' = \mathbf{A} + \nabla\Lambda, \quad \psi \mapsto \psi' = e^{i(q/\hbar)\Lambda}\psi,$$

where

$$\Lambda(\mathbf{r}) \equiv -\int_{\mathbf{r_0}}^{\mathbf{r}} \mathbf{A}(\mathbf{r}')\cdot d\mathbf{r}'$$

such that $\nabla\Lambda = -\mathbf{A}$ and $\mathbf{A}' = 0$. Note that the "gauge potential" Λ is independent of the path taken from some reference point $\mathbf{r}_0$ to the point $\mathbf{r}$ since $\nabla \times \mathbf{A} = 0$, *assuming the space is simply connected.* Since we've effectively removed the vector potential, the Schrödinger equation becomes

$$i\hbar\frac{\partial\psi'}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi'$$

which says that ψ' satisfies the Schrödinger equation without the vector potential $\mathbf{A}$. As it turns out, there are circumstances in which the gauge potential cannot be globally defined, and $\mathbf{A}$ cannot simply be removed.

In order for Λ to be a well-defined scalar function, we need the line integral of $\mathbf{A}$ to be zero for every closed loop. In a simply connected region, the condition $\nabla \times \mathbf{A} = 0$ implies exactly that according to Stokes' theorem. To see where the argument above fails, consider two different paths, C_1 and C_2, from $\mathbf{r}_0$ to $\mathbf{r}$ and define $\Lambda_1(\mathbf{r})$ along C_1 and $\Lambda_2(\mathbf{r})$ along C_2. Their difference is given by

$$\Lambda_1(\mathbf{r}) - \Lambda_2(\mathbf{r}) = -\oint_C \mathbf{A}(\mathbf{r}')\cdot d\mathbf{r}'$$

where C is a closed loop formed by going out along C_1 and back along C_2. Applying Stokes' theorem, we find that the difference is related to the magnetic flux:

$$\oint_C \mathbf{A}(\mathbf{r}')\cdot d\mathbf{r}' = \int_S (\nabla\times\mathbf{A})\cdot d\mathbf{a} = \int_S \mathbf{B}\cdot d\mathbf{a} = \Phi_B$$

where S is a surface bounded by the closed curve C. Whenever the magnetic flux is nonzero, the difference between Λ_1 and Λ_2 will be nonzero, and the function cannot be well-defined.

For example, consider a particle that is confined to move on a ring of radius a surrounding a solenoid of radius $b < a$. If the solenoid is extremely long, the magnetic field inside it is uniform, and the field outside is zero. Let $\Lambda_1(\theta)$ correspond to a path taken from a reference line where $\theta = 0$ to the angle θ around the ring and $\Lambda_2(\theta)$ correspond to a path taken from $\theta = 0$ all the way around the ring, and then to θ. Geometrically, we have

$$\int_{C_2} \mathbf{A}(\mathbf{r}')\cdot d\mathbf{r}' = \oint_C \mathbf{A}(\mathbf{r}')\cdot d\mathbf{r}' + \int_{C_1} \mathbf{A}(\mathbf{r}')\cdot d\mathbf{r}',$$

or

$$\Lambda_2(\mathbf{r}) = \Lambda_1(\mathbf{r}) - \Phi_B$$

Using single-valuedness of the wave function,

$$\psi_2'(\theta+2\pi) = e^{i(q/\hbar)\Lambda_2}\psi(\theta+2\pi) = e^{i(q/\hbar)(\Lambda_1-\Phi_B)}\psi(\theta) = e^{-i(q/\hbar)\Phi_B}\psi_1'(\theta)$$

which can be thought of as a "twisted boundary condition" in the sense that ψ_1' and ψ_2' are both solutions to the same free-particle Schrödinger equation, but the nonzero magnetic flux enclosed by the loop leads to an overall phase difference as the ring is encircled. We can easily determine the energy eigenvalues using the free-particle solution $\psi' = e^{ik\theta}$[1]. Imposing the twisted boundary condition,

$$e^{ik(\theta+2\pi)} = e^{-(q/\hbar)\Phi_B} e^{ik\theta} \Rightarrow k = n - \frac{q\Phi_B}{2\pi\hbar},\ n \in \mathbb{Z}$$

Therefore, the energy eigenvalues are

$$E_n = \frac{\hbar^2}{2I}\left(n - \frac{q\Phi_B}{2\pi\hbar}\right)^2$$

so the effect of the flux is simply to shift the integer quantum number $n \to n - \Phi_B/\Phi_0$. The solenoid lifts the two-fold degeneracy of the particle-on-a-ring: positive n, representing a particle traveling in the *same direction* as the current in the solenoid, has a somewhat lower energy (assuming q is positive) than negative n, describing a particle traveling in the *opposite direction*. More importantly, the allowed energies clearly depend on the field inside the solenoid, even though the field at the location of the particle is zero!

Aharonov and Bohm proposed an experiment in which a beam of electrons is split in two, and they pass either side of a long solenoid before recombining. The beams are kept well away from the solenoid itself, so they encounter only regions where $\mathbf{B} = 0$. The magnetic flux contained within a closed path around the solenoid is not zero, and the two beams arrive with different phases:

$$\text{phase difference} \ \propto \ \frac{\Phi_B}{\Phi_0}$$

This phase shift leads to measurable interference, which has been confirmed experimentally. Another consequence of the Aharonov–Bohm (AB) effect was the discovery that magnetic flux is likely not quantized; if the total magnetic flux were quantized in integer multiples of Φ_0, no interference pattern would be observed.

The AB effect is called a *topological effect*, in that the effect depends on the flux encircled by the paths available to the particle, even though the paths may never approach the region of the flux. Since the magnetic force on the charge vanishes on all possible paths of the particle, one might wonder whether the charge is necessary for the effect. According to the theory, the effect depends on the dimensionless ratio $q\Phi_B/h$, which is proportional to the charge, and it has been experimentally confirmed that no AB effect occurs if neutrons are used instead of electrons.

[1] See Chapter #8, *Particle on a Ring*